

Xuanming Zhang

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Research Interests

Static, rule-based AI alignment fails to navigate the dynamic and often contradictory nature of human values. My research introduces Humanoid Intelligence as a new paradigm, aiming to build AI that intrinsically reasons about social context rather than merely following predefined rules. The ultimate goal is to create AI that can deeply understand and carry forward human will, enabling future applications like the digital continuity of consciousness. My work is concentrated in these core areas:

- **Cognitive Learning:** Building trustworthy AI that can generalize its understanding of human values to novel situations. My work develops cognitive architectures including test-time learning mechanisms and metacognitive multi-agent system that enable case-by-case reasoning, moving beyond the limitations of fixed training data to achieve robust, adaptable alignment.
- **World Interaction:** Developing agents that learn through immersive interaction within simulated, long-horizon social scenarios. This allows for the study and shaping of complex emergent behaviors like deception, cooperation, and contextual decision-making in a controlled yet realistic setting.
- **Policy Simulation:** Conducting evaluation by forecasting the long-term societal impact of an agent's decisions. This shifts the measure of success from immediate, task-specific correctness to the broader, downstream consequences of AI behavior in complex systems.

Education

B. S. in Information Science Jan. 2025 May. 2026
[University of Wisconsin-Madison](#), Madison, WI
• Overall GPA: 3.84/4.0.

Visiting B. S. in Computer Science Jun. 2025 Aug. 2025
[Stanford University](#), Stanford, CA
• Overall GPA: A+.

B.S. in Artificial Intelligence Sept. 2021 Dec. 2023
Dropout from the Program of National Elite Institute of Engineering
• Outstanding Student (Top 1).
• First-Class Scholarship.

Awards and Certificate

- National Level II Psychological Counselor, PCIA, 2025.
- The Chinese Mathematics Competitions for College Students, First Prize, 2023.
- National College Student Innovation and Entrepreneurship Training Program (\$10,000), Ministry of Education in China, 2023.

Publications

See full list in [my google scholar](#) page, I publish under the name "Xuanming Zhang".
* indicates Equal Contribution

11. [Cognition-of-Thought Elicits Social-Aligned Reasoning in Large Language Models](#)
Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
 International Conference on Learning Representations (ICLR) under review, 2026.
10. [Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions](#)
 Yang Xu*, **Xuanming Zhang***, Min-Hsuan Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Yixuan Li
 International Conference on Learning Representations (ICLR) under review, 2026.
9. [MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems](#)
Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
 Advances in Neural Information Processing Systems (NeurIPS) 2025, **spot-light**.
8. [Human Decision-making is Susceptible to AI-driven Manipulation](#)
 Sahand Sabour, June M Liu, Siyang Liu, Chris Z Yao, Shiyao Cui, **Xuanming Zhang**, Wen Zhang, Yaru Cao, Advait Bhat, Jian Guan, Wei Wu, Rada Mihalcea, Hongning Wang, Tim Althoff, Tatia Lee, Minlie Huang
 Nature Communications under revision, 2025.
7. [SocialEval: Evaluating Social Intelligence of Large Language Models](#)
 Jinfeng Zhou, Yuxuan Chen, Yihan Shi, **Xuanming Zhang**, Leqi Lei, Yi Feng, Zexuan Xiong, Miao Yan, Xunzhi Wang, Yaru Cao, Jianing Yin, Shuai Wang, Quanyu Dai, Zhenhua Dong, Hongning Wang, Minlie Huang
 Annual Meeting of the Association for Computational Linguistics (ACL) 2025.
6. [Towards Exception Safety Code Generation with Intermediate Representation Agents Framework](#)
Xuanming Zhang, Yuxuan Chen, Yuan Yuan, Minlie Huang
 ACM Transactions on Software Engineering and Methodology (TOSEM) under revision, 2025.
5. [Deep Graph Learning for Industrial Carbon Emission Analysis and Policy Impact](#)
Xuanming Zhang
 Advances in Neural Information Processing Systems (NeurIPS) 2025, AI for Science Workshop.
4. [EvoCodeBench: An Evolving Code Generation Benchmark with Domain-specific Evaluations](#)
 Jia Li, Ge Li, **Xuanming Zhang**, Yunfei Zhao, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
 Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024.
3. [DevEval: A Manually-Annotated Code Generation Benchmark Aligned with Real-World Code Repositories](#)
 Jia Li, Ge Li, Yunfei Zhao, Yongmin Li, Huanyu Liu, Hao Zhu, Lecheng Wang, Kaibo Liu, Zheng Fang, Lanshen Wang, Jiazheng Ding, **Xuanming Zhang**, Yuqi Zhu, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
 Annual Meeting of the Association for Computational Linguistics (ACL) 2024.

Preprints	2. Theoretical Proof that Auto-regressive Language Models Collapse when Real-world Data is a Finite Set Lecheng Wang, Xianjie Shi, Ge Li, Jia Li, Xuanming Zhang , Yihong Dong, Wenpin Jiao, Hong Mei
	1. Generalization or Memorization: Dynamic Decoding for Mode Steering Xuanming Zhang
Service	<i>Reviewer for</i> • NeurIPS (Top Reviewer), ICLR • OOPSLA • AAAI • TMLR
	<i>Mentees</i> 2. Tingting Du , undergraduate student at UW-Madison, 2025. 1. Yuxuan Chen , undergraduate student at THU, 2024, now MS student at THU.
Research Experience	<i>Student Researcher</i> May 2025 - Sep. 2025 Amazon, Remote, USA • Hosts: Dr. Rahul Gupta, Jwala Dhamala and Ousmane Dia. • Research on <i>Trustworthy AI: Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions</i>
	<i>Research Assistant</i> Feb. 2025 - May. 2026 Department of Computer Sciences, UW-Madison, Madison, WI, USA • Supervised by Dr. Sharon Li. • Research on <i>Socially aligned LLMs</i>.
	<i>AI Architect Intern</i> Sep. 2024 - Jan. 2025 Flow, Bytedance, Beijing, China • Supervised by Dr. Xingnan Wang and Xuebin Lin. • Research and design on Coze.
	<i>Research Scientist Intern</i> Apr. 2024 - Jan. 2025 Department of Computer Science, Tsinghua University & Z.ai, Beijing, China • Supervised by Dr. Minlie Huang. • Research on <i>AI alignment, safety and ethics</i>.
	<i>Member</i> Jun. 2023 - May. 2024 Key Lab of High Confidence Software Technologies, Ministry of Education & School of Computer Science, Peking University, Beijing, China • Supervised by Dr. Zhi Jin and Dr. Ge Li. • Research on <i>Controlled-Text Generation</i>.
	<i>Language skills</i> • Native speakers of Mandarin with professional English speaking capability .
Additional Information	<i>Programming skills</i> • Proficient with Python, PyTorch. Familiar with Matlab, C++, Swift.

Leadership skills

- I worked as CTO at the startup H3O from 2021-2022 and secured \$70,000 in funding.
- I was the president of the student union both in BNDS and in college.